Week 3 Report

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Role: Frontend Developer & Scrum Master

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Project: OAS - Web-based Oral Assessments

Reporting Period: Monday, August 11, 2025 – Sunday, August 17, 2025

1. Tasks Completed This Week

Frontend Development & UI/UX Improvements

Authentication System Integration & Enhancement

- **Spring Boot Backend Integration** (Evidence Link)
  - Configured the API client to connect with the Spring Boot backend.
  - Implemented JWT token management, including accessToken and refreshToken.
  - Mapped backend roles (e.g., ROLE_ADMIN) to frontend roles.
  - Updated the authentication service to use backend endpoints for login, logout, and token refresh.
  - Connected logout functionality to the backend for proper token invalidation.

Comprehensive Examiner Dashboard Implementation (Evidence Link)

- **Dashboard Infrastructure:**
  - Created a main layout for the examiner view with a sidebar and top navigation.
  - Implemented a hierarchical navigation menu in the sidebar.
  - Developed a top navigation bar with search, notifications, and a user menu.
- **Dashboard Components:**
  - Designed cards to display key metrics (e.g., assessments, feedback).
  - Created a "Today's Schedule" component with action buttons.
  - Implemented a "Pending Feedback" section with priority indicators.
  - Added a compact weekly calendar with color-coded status indicators.
- **Feature Pages:**
  - Developed interfaces for assessments, a comment library, schedule

management, results, and profile settings.

Unified Onboarding System (Evidence Link)

- **Multi-Role Onboarding Flow:**
  - Implemented a complete token-based onboarding process for Course Admins, Students, and Examiners.
  - Created a role-aware UI with differentiated steps and themes.
  - Implemented secure password creation with real-time validation.
- **Testing & Integration:**
  - Developed a comprehensive mock API with multiple test scenarios.
  - Added graceful error handling for expired or invalid tokens.

Password Reset System (Evidence Link)

- **Security-First Implementation:**
  - Created a "Forgot Password" page with an email submission form.
  - Implemented a "Reset Password" page with 6-digit OTP verification.
  - Added password complexity validation with real-time feedback and a resend timer.
- **Security Features:**
  - Ensured generic success messages to prevent account enumeration.
  - Implemented attempt tracking and validation to ensure the new password differs from the old one.

Scrum Master Responsibilities

- **Sprint Planning & Coordination:** Coordinated the Week 3 sprint plan, facilitated weekly standups, and managed the integration timeline between frontend and backend teams.
- **Technical Leadership & Code Quality:** Conducted code reviews for new features, ensured architectural consistency, and established testing protocols for authentication flows.
- **Documentation & Knowledge Management:** Created comprehensive API documentation for backend integration and updated the project README.

2. Time Allocation (Approximation for Week 3)

- **Frontend Development:** 28 hours
  - Authentication Integration: 8 hours
  - Examiner Dashboard: 8 hours
  - Onboarding System: 6 hours
  - Password Reset: 6 hours
- **Scrum Master Activities:** 3 hours

    - Sprint planning and coordination: 1 hour
    - Daily standups and team communication: 1 hour
    - Code reviews and technical guidance: 1 hour
- **Documentation & Testing:** 4 hours
    - API documentation creation: 2 hours
    - Testing and validation: 2 hours

Total: 35 hours

3. Key Issues Encountered & Resolutions/Mitigation

Backend Integration Challenges

- **Issue:** The initial authentication flow required significant refactoring to align with the Spring Boot JWT implementation.
- **Resolution:** Redesigned the authentication store to handle the backend's response structure, including proper token management and role mapping.
- **Mitigation:** Created comprehensive test accounts and documented the integration process to serve as a reference for future work.

Component Architecture Complexity

- **Issue:** The Examiner Dashboard implementation required careful integration with the existing authentication system to avoid conflicts.
- **Resolution:** Maintained strict compatibility by preserving existing authentication patterns while adding new dashboard functionality.
- **Mitigation:** Implemented a modular component architecture to prevent breaking changes to existing features.

Onboarding Flow State Management

- **Issue:** The complex, multi-step onboarding process required sophisticated state management to handle different user roles.
- **Resolution:** Implemented role-aware UI components with centralized state management and robust error handling.
- **Mitigation:** Created a dedicated testing interface to validate all onboarding scenarios before backend integration.

4. Plan for Next Week (Week 4: August 18 - August 24, 2025)

Priority 1: Backend Integration Completion

- Complete integration testing of the authentication system with the Spring Boot backend.
- Validate the password reset flow with the actual email service integration.

- Test the onboarding system with real tokens generated from the backend.

Priority 2: Student Dashboard Development

- Implement a comprehensive student dashboard using the examiner dashboard as a pattern.
- Create student-specific components (e.g., upcoming assessments, results).
- Develop the assessment-taking interface with real-time features.

Priority 3: Course Admin Interface Enhancement

- Expand the course admin dashboard with assessment management features.
- Implement workflows for course creation and management.
- Add interfaces for student enrollment and examiner assignment.

Priority 4: Testing & Quality Assurance

- Conduct end-to-end testing of all authentication flows.
- Perform cross-browser compatibility testing and validate accessibility compliance.
- Complete final integration testing with the backend team.

Scrum Master Focus Areas

- Coordinate the final sprint planning for the authentication system.
- Facilitate integration testing sessions between frontend and backend teams.
- Manage the resolution of technical debt and plan user acceptance testing (UAT).